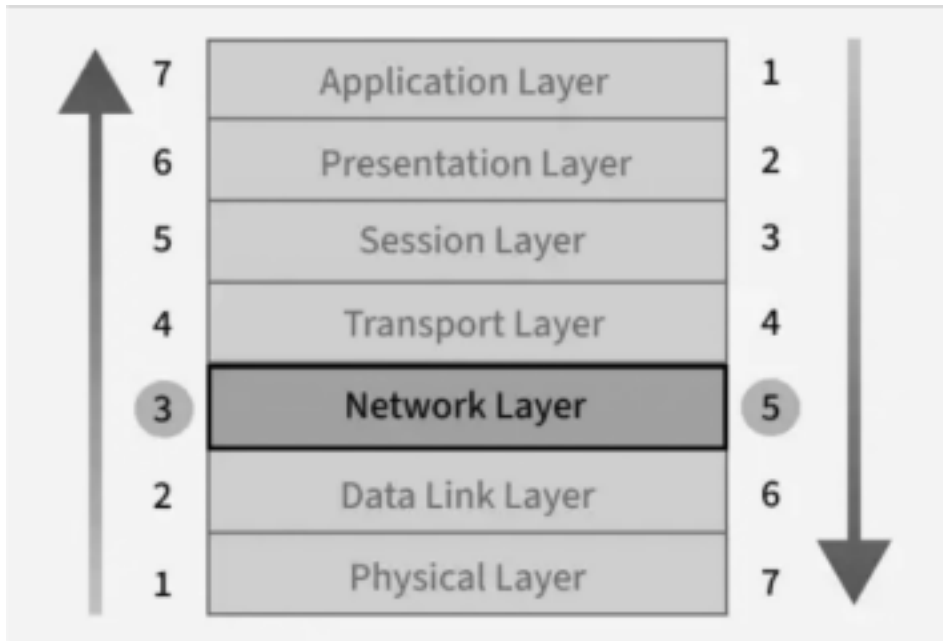


Network Layer

1. What is the Network Layer?

The **Network Layer** is the **3rd layer** in the **OSI (Open Systems Interconnection) model**. It sits between the Data Link Layer (Layer 2) and the Transport Layer (Layer 4).



Purpose

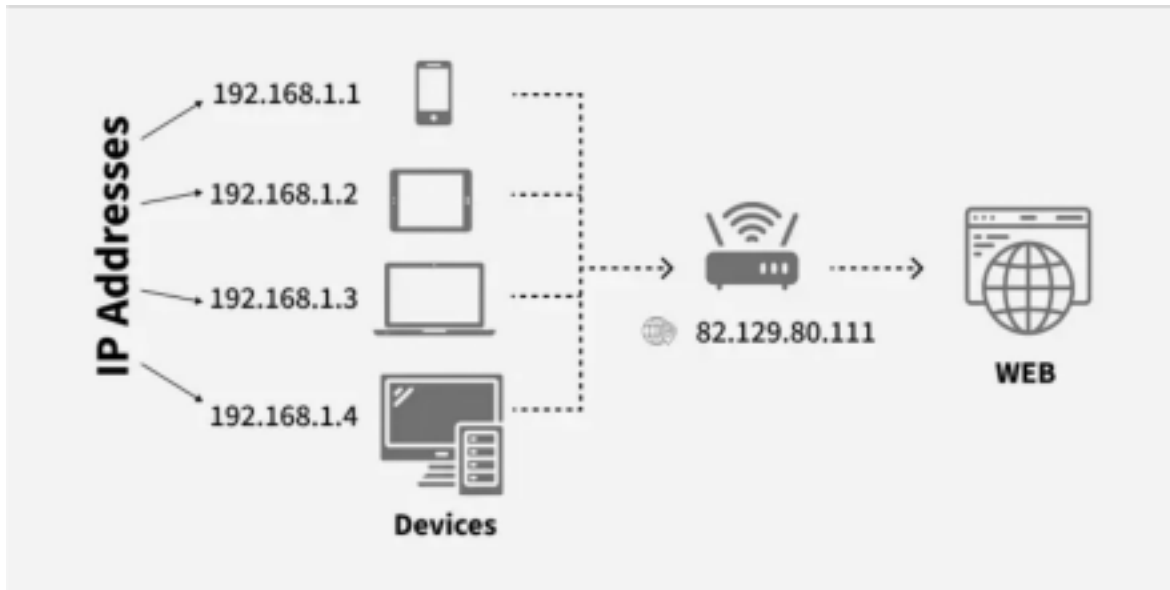
- Enables **communication between different networks** (internetworking).
- It is responsible for **logical addressing, routing, packet forwarding**, and **end-to-end delivery** of data.

Key Functions

- Logical addressing (assigns IP addresses)
- Packet formation and forwarding
- **Routing** through optimal paths
- **Subnetting**, dividing bigger networks into smaller groups
- **Fragmentation & reassembly** of packets
- Interconnecting different networks
- Handling **Network Address Translation (NAT)**
- Working with protocols such as IP, ICMP, ARP, etc.

2. What is an IP Address?

An **IP (Internet Protocol) address** is a **unique numeric label** assigned to every device on a network so it can send and receive data packets.



Components of an IP Address

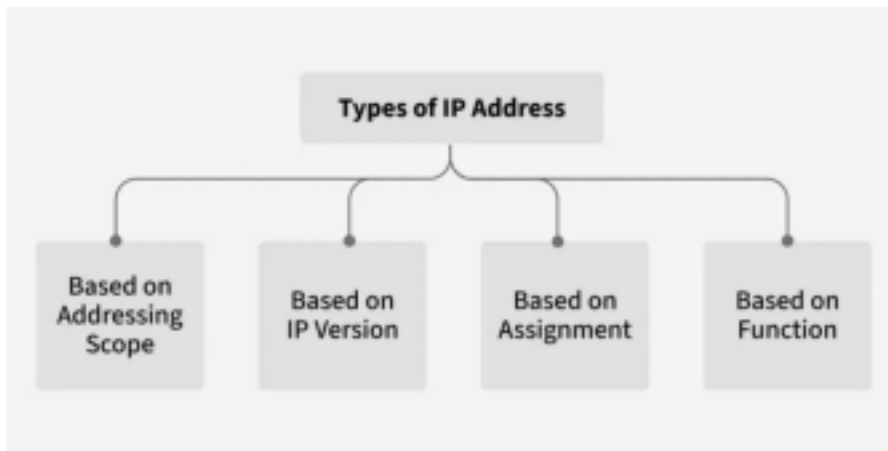
1. **Network Portion:** Identifies the network to which the device belongs.
2. **Host Portion:** Identifies the individual device on the network.
3. **Subnet Mask (for IPv4):** Defines which part of the IP is network and which part is host.

Example: IP 192.168.1.10 with subnet mask 255.255.255.0

Network ID: 192.168.1.0

Host ID: 10

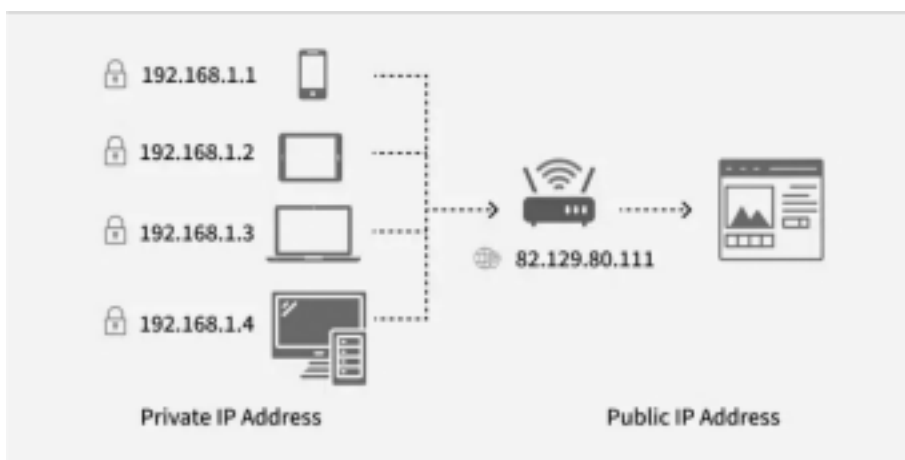
Types of IP Address



1. Based on Addressing Scope.

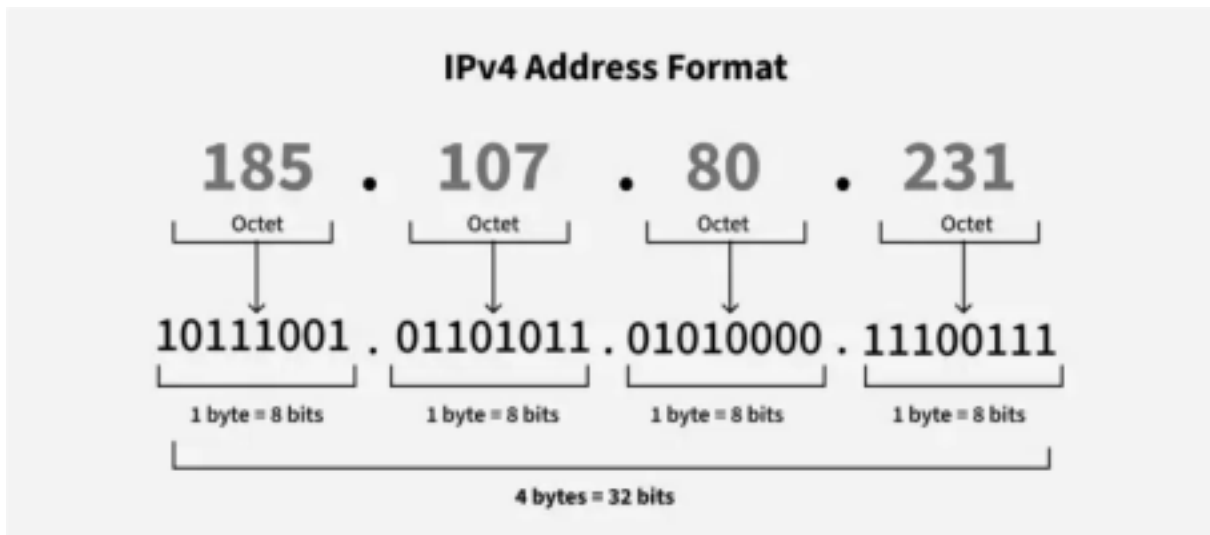
Public IP Addresses: A Public IP address is assigned to every device that directly accesses the internet. This address is unique across the entire internet. Uniqueness & Accessibility are its key characteristics & are assigned by Internet Service Providers. When you connect to the internet through an ISP, your device or router receives a public IP address. These addresses can be static or dynamic.

Private IP Addresses: Private IP addresses are used within private networks and are not routable on the internet. This means that devices with private IP addresses cannot directly communicate with devices on the internet without a translating mechanism like a router performing Network Address Translation (NAT). These are only required to be unique within their own network & are used for communication between devices within the same network

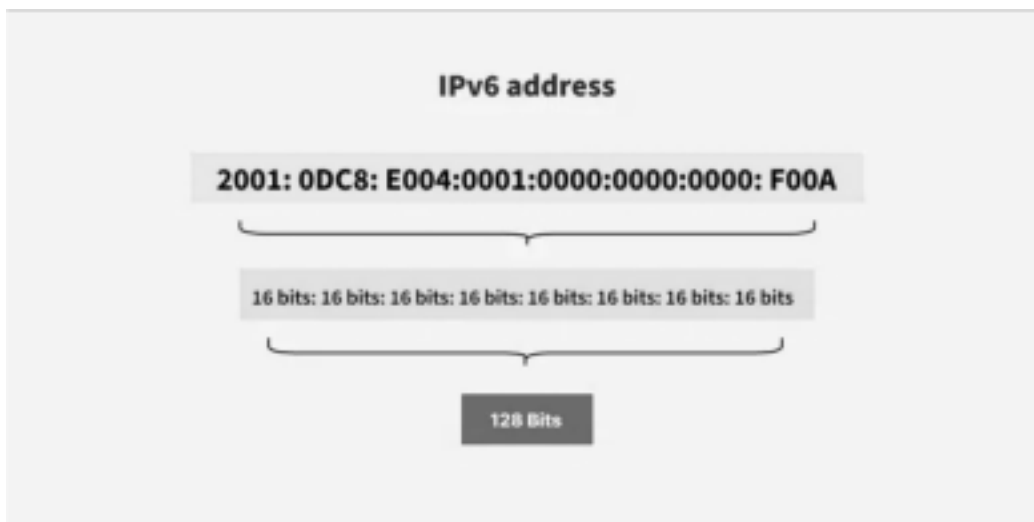


2. Based on IP Version (IPv4 vs. IPv6)

IPv4: This is the most common form of IP Address. It consists of four sets of numbers(octets) separated by dots. This format can support over 4 billion unique addresses. Each octet represents eight bits, or a byte, and can take a value from 0 to 255. This range is derived from the possible combinations of eight bits ($2^8 = 256$ combinations). Can support about **4.3 billion** unique addresses.



IPv6: IPv6 addresses were created to deal with the shortage of IPv4 addresses. They use 128 bits instead of 32, offering a vastly greater number of possible addresses. These addresses are expressed as eight groups of four hexadecimal digits, each group representing 16 bits. The groups are separated by colons. Designed to overcome IPv4 address exhaustion.



3. Based on Assignment

Static IP Addresses: Static IP Addresses are permanently assigned to a device, typically important for servers or devices that need a constant address. Reliable for network services that require regular

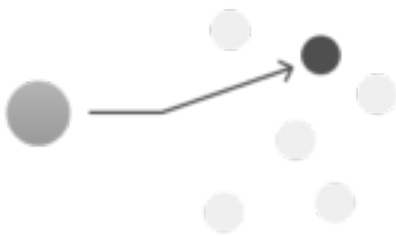
access such as websites, remote management.

Dynamic IP Addresses: Temporarily assigned from a pool of available addresses by the Dynamic Host Configuration Protocol (DHCP). Cost-effective and efficient for providers, perfect for consumer devices that do not require permanent addresses.

4. Based on Function

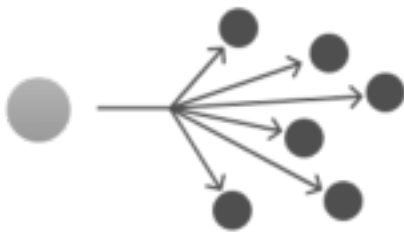
Unicast Address: In unicast, data is sent from one sender to one specific receiver identified by a unique IP address. It is the most common type of communication used in networks. Its purpose is One-to-one communication.

Use Case: Regular web browsing, file transfers (FTP), email (SMTP), etc.



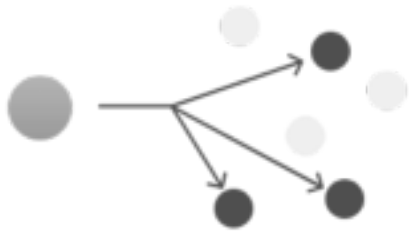
Broadcast Address: In broadcast, a message is sent from one device to all devices in the same network segment. Every device in the network receives and processes the broadcast message. Its Purpose is One-to-all communication within a network.

Use Case: Network discovery, DHCP requests, ARP queries



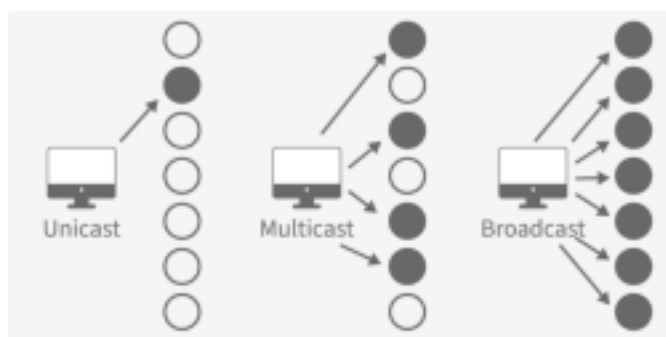
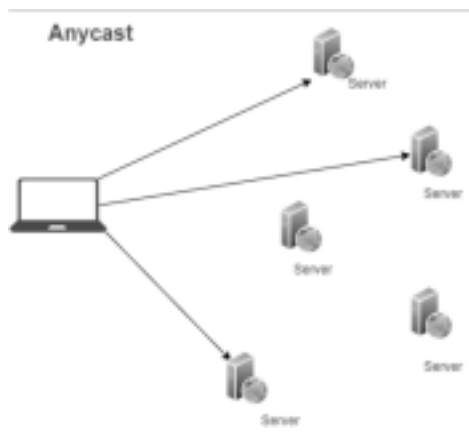
Multicast Address: In multicast, data is sent from one source to multiple selected receivers that are part of a multicast group. Only devices that have joined the group will receive the data, making it more efficient than broadcasting. Its Purpose is One-to-many (selected group) communication.

Use Case: video conferencing, live streaming



Anycast Address: In anycast, data is sent from one sender to the nearest receiver (in terms of network distance) among a group of devices sharing the same IP address. Routers determine the closest destination dynamically. Its Purpose is One-to-nearest communication (based on routing distance).

Use Case: DNS servers, CDN routing, load balancing



4. Introduction of Classful IP Addressing

Classful IP addressing was an early method (1981–1993) used to assign IP addresses and divide the IPv4 address space into fixed classes. Each IP address was categorized based on its leading bits, and the class determined its network and host portions. Due to inefficient address utilization, this method was later replaced by CIDR, which uses flexible prefixes instead of fixed classes.

Classes & Their Purpose:

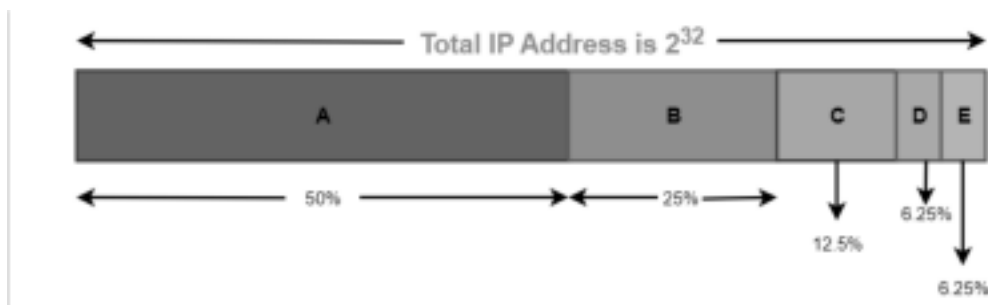
- IP addresses were divided into five classes: A, B, C, D, and E.
- The class of an IP address was identified by its first few bits.
- Classes A, B, and C were used for unicast communication, supporting large, medium, and small networks, respectively.
- Class D was reserved for multicast communication.
- Class E was intended for experimental and research purposes.

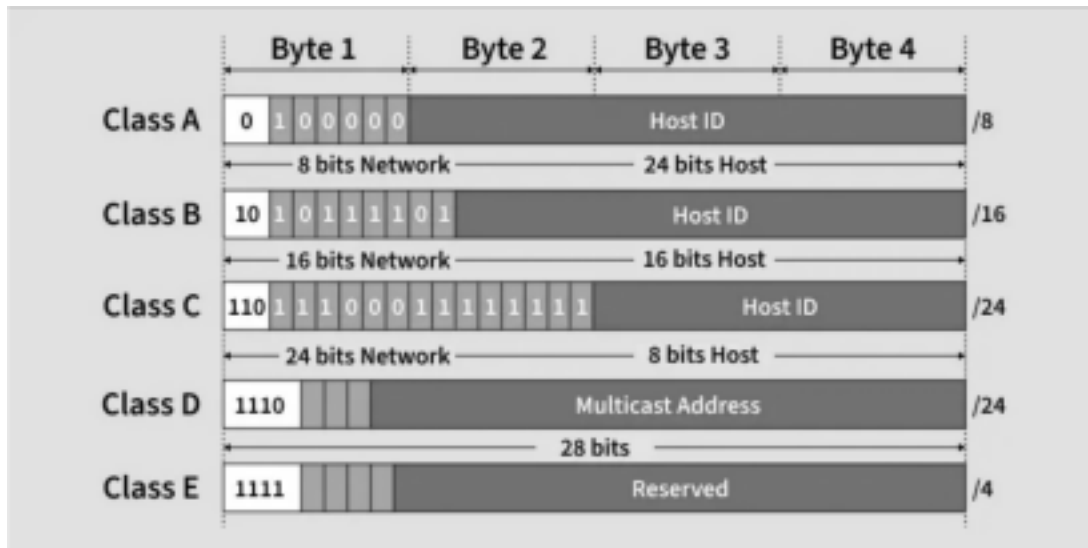
IP Classes

Class	Range	Default Mask	Purpose
A	1.0.0.0 - 126 .255.255.255	/8	Large networks
B	128 .0.0.0 - 191 .255.255.255	/16	Medium networks
C	192 .0.0.0 - 223 .255.255.255	/24	Small networks
D	224 .0.0.0 - 239 .255.255.255	N/A	Multicast
E	240 .0.0.0 - 255 .255.255.255	N/A	Experimental

Classful addressing is **static and inefficient**, often causing large unused address spaces. which is why it was replaced later.

Occupation of The Address Space In Classful Addressing





1. Network address: Identifies the network itself (all host bits = 0)

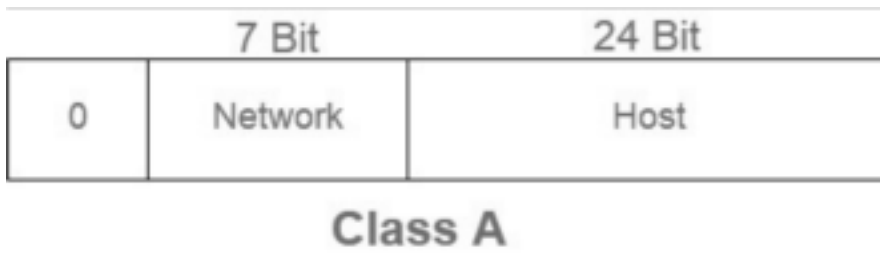
Example: 192.168.1.0

2. Broadcast address: Used to send to all hosts on that network (all host bits = 1)

Example: 192.168.1.255

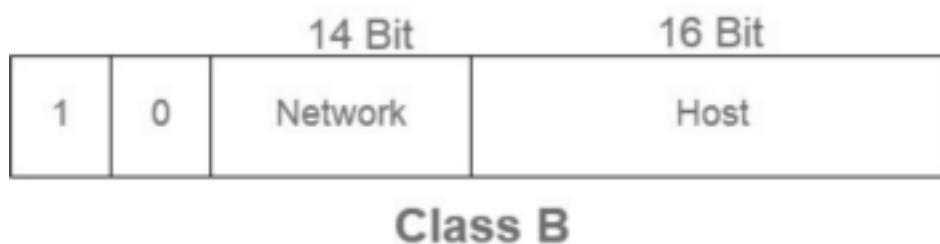
Class A

- Class A IP addresses are assigned to networks that require a very large number of hosts.
- The network ID in Class A is 8 bits long, while the host ID is 24 bits long.
- The most significant bit (MSB) of the first octet is always 0, and the remaining 7 bits are used to identify the network.
- The 24-bit host ID uniquely identifies hosts within the same network.
- The default subnet mask for Class A is 255.0.0.0.
- The total number of usable host addresses per network is $2^{24} - 2 = 16,777,214$ (excluding network and broadcast addresses).
- The IP address range for Class A is 0.0.0.0 to 127.255.255.255.



Class B

- Class B IP addresses are assigned to medium to large-sized networks.
- The network ID is 16 bits long, and the host ID is also 16 bits long.
- The first two bits of the first octet are always 10, which identifies a Class B address.
- The remaining 14 bits are used to determine the network ID.
- The 16-bit host ID uniquely identifies hosts within the same network.
- The default subnet mask for Class B is 255.255.0.0.
- The total number of Class B networks is $2^{14} = 16,384$ networks.
- The total number of usable host addresses per network is $2^{16} - 2 = 65,534$ hosts.
- The IP address range for Class B is 128.0.0.0 to 191.255.255.255.



Class C

- Class C IP addresses are assigned to small-sized networks.
- The network ID is 24 bits long, while the host ID is 8 bits long.
- The first three bits of the first octet are always 110, which identifies a Class C address.
- The remaining 21 bits are used to determine the network ID.

- The 8-bit host ID uniquely identifies hosts within the same network.
- The default subnet mask for Class C is 255.255.255.0.
- The total number of Class C networks is $2^{21} = 2,097,152$ networks.
- The total number of usable host addresses per network is $2^8 - 2 = 254$ hosts.
- The IP address range for Class C is 192.0.0.0 to 223.255.255.255.



Class D

- Class D IP addresses are reserved for multicast communication.
- The first four bits of the first octet are always 1110, which identifies a Class D address.
- The remaining 28 bits are used to represent the multicast group address that interested hosts can join.
- Class D addresses do not have a network ID or host ID division.
- No subnet mask is defined for Class D addressing.
- The IP address range for Class D is 224.0.0.0 to 239.255.255.255.



Class E

- Class E IP addresses are reserved for experimental and research purposes.
- The first four bits of the first octet are always 1111, which identifies a Class E address.

Class E addresses do not have network ID and host ID divisions.

- No subnet mask is defined for Class E addressing.
- The IP address range for Class E is 240.0.0.0 to 255.255.255.255.



Range of Special IP Addresses

- **169.254.0.0 – 169.254.255.255**

Used as link-local addresses when a device cannot obtain an IP address from a DHCP server.

- **127.0.0.0 – 127.255.255.255**

Reserved for loopback addresses, used to test network functionality on the local machine.

- **0.0.0.0 – 0.255.255.255 (0.0.0.0/8)**

Represents the current network and is used during initialization before a device is assigned a valid IP address.

Rules for Assigning Host ID

Host IDs are used to uniquely identify a host within a network and must follow these

rules: ● **Uniqueness:** Each host ID must be unique within the same network.

- **All-zeros restriction:** A host ID with all bits set to 0 is not allowed, as it represents the network address.

- **All-ones restriction:** A host ID with all bits set to 1 is not allowed, as it is reserved for the broadcast address used to reach all hosts in the network.

Structure of Classful Addressing



Problems With Classful Addressing

Classful addressing resulted in inefficient IP address utilization due to fixed address classes. Class A and Class B networks wasted a large number of addresses, while Class C networks offered too few hosts to support growing organizations. Class D addresses were limited to multicast use, and Class E addresses were reserved for experimentation. Because of this inflexibility and poor scalability, classful addressing was replaced by CIDR in 1993.

5. Classless Addressing in IP Addressing

Classless Addressing is also called **CIDR (Classless Inter-Domain Routing)**. It replaced classful addressing because of IP address inefficiency.

Classless Details

- There are **no fixed classes**.
- IP looks like: **192.168.1.0/28** - where **/28** tells how many bits are for the network.
- This lets networks create **exactly needed sizes** without wasting IPs.
- Supports **supernetting** and **route aggregation**.

Example: 192.168.1.0/28

1. Given IP:

192.168.1.0/28

/28 means:

- 28 bits = Network bits
- 4 bits = Host bits ($32 - 28 = 4$)

2. Subnet Mask:

/28 in decimal form: 255.255.255.240

Binary: 11111111.11111111.11111111.11110000

3. Number of Hosts

Formula: $2^{(\text{host bits})} - 2$

$2^4 - 2 = 16 - 2 = 14$ usable hosts

So this subnet can support **14 devices**.

4. Address Range

Block size formula: $256 - 240 = 16$

So the subnet increases by **16**.

For 192.168.1.0/28:

- Network Address: 192.168.1.0
- First Host: 192.168.1.1
- Last Host: 192.168.1.14

- Broadcast Address: 192.168.1.15

5. Next Subnet

Since block size = 16:

Next subnet will be: 192.168.1.16/28

Then:

192.168.1.32/28

192.168.1.48/28

Final

For **192.168.1.0/28**:

- Subnet Mask: **255.255.255.240**
- Total IPs: **16**
- Usable Hosts: **14**
- Network Address: **192.168.1.0**
- Broadcast Address: **192.168.1.15**
- Valid Host Range: **192.168.1.1 – 192.168.1.14**

Benefits of Classless Addressing

- Efficient utilization of IP space
- Routing tables are smaller
- Supports **Variable Length Subnet Masks (VLSM)**
- Better scalability

6. IPv4 Datagram Header

Each IPv4 packet has a header that includes information needed to route and deliver the packet.



IPv4 Datagram Header

Minimum size = **20 bytes**

Maximum size = **60 bytes** (if options are used)

1. Version (4 bits)

- Tells which IP version is used.
- For IPv4, the value is **4**.
- Routers check this field first.

✓ Example: If value = 6 → It is an IPv6 packet.

2. HLEN (Header Length) – 4 bits

- Also called **IHL (Internet Header Length)**.
- Tells the length of the header in **32-bit words**.
- Minimum value = **5** → (5×4 bytes = 20 bytes)
- Maximum value = **15** → (15×4 bytes = 60 bytes)

3. Type of Service (8 bits)

- Indicates the quality of service required.
- It helps routers decide packet priority.

Earlier divided into:

- Low Delay
- High Throughput
- High Reliability

Now it is called **DSCP (Differentiated Services Code**

Point). Different types of traffic need different priorities:

- Video calls → Need low delay
- Voice calls → Need very low delay
- Email → Can tolerate delay
- File download → Can tolerate delay

DSCP helps routers treat traffic differently.

4. Total Length (16 bits)

- Total size of the packet (Header + Data).
- Minimum = 20 bytes (header only)
- Maximum = 65,535 bytes

Example:

Header = 20 bytes

Data = 1000 bytes

Total Length = 1020 bytes

5. Identification (16 bits)

- Unique number assigned to each packet.
- Used when a packet is fragmented.
- All fragments of the same packet have the same Identification number.

Helps in reassembling fragments at destination.

6. Flags (3 bits)

Used for fragmentation control:

1. Reserved bit → Always 0
2. DF (Do Not Fragment) → If 1, packet cannot be fragmented
3. MF (More Fragments) → If 1, more fragments follow

The last fragment has MF = 0.

7. Fragment Offset (13 bits)

- Tells the position of a fragment in the original packet.
- Measured in units of **8 bytes**.
- Maximum value = 65,528 bytes.

✓ Used to correctly reassemble fragmented packets.

8. Time To Live (TTL) – 8 bits

- Limits the life of a packet.
- Prevents infinite looping in the network.
- Decreased by 1 at each router (hop).
- When TTL = 0 → Packet is discarded.

Example: TTL = 64 → Packet can travel 64 routers.

9. Protocol (8 bits)

- Tells which upper-layer protocol will receive the data.

Examples:

- 1 → ICMP

- 6 → TCP
- 17 → UDP

✓ Helps in delivering data to correct applications.

10. Header Checksum (16 bits)

- Used to detect errors in header only (not data).
- Checked at every router.
- If error found → Packet discarded.

11. Source IP Address (32 bits)

- IP address of sender.
- Example: 192.168.1.10

12. Destination IP Address (32 bits)

- IP address of receiver.
- Example: 8.8.8.8

13. Options (Variable Length)

- Optional field.
- Used for special purposes like:
 - Source routing
 - Record route
 - Network testing

✓ If options are present → Header length increases.

9. Introduction to Subnetting

Subnetting is dividing a large IP network into **smaller sub-networks** (subnets) for better management and efficiency.

1. Reason for Subnetting

Subnetting is used to:

✓ **Efficient IP Address Usage:** Instead of wasting IP addresses, we divide a large network into smaller usable networks.

✓ **Improve Security:** Departments (HR, Sales, IT) can be separated

logically. ✓ **Better Network Management**

2. What is Subnetting?

Subnetting is the process of dividing one large IP network into multiple smaller networks (subnets).

Example:

If you have a network:

192.168.1.0/24

It has **256 total addresses**.

You can divide it into smaller networks like:

192.168.1.0/26

192.168.1.64/26

192.168.1.128/26

192.168.1.192/26

Now you have **4 smaller networks**, each with 64 addresses.

3. Types of Subnetting

1. Fixed Length Subnetting (FLSM)

All subnets are the **same size**.

You borrow the same number of bits for every subnet.

Network having IP Address 200.1.2.0. Divide this network into two subnets.

1st Subnet

- IP Address of the subnet / Subnet id = 200.1.2.0
- Direct Broadcast Address = 200.1.2.127
- Total number of IP Addresses = 128
- Range of IP Addresses = [200.1.2.0, 200.1.2.127]
- Total number of hosts that can be configured = 128 - 2 = 126
- Range of Allocated IP Addresses = [200.1.2.1, 200.1.2.126]

2nd Subnet

- IP Address of the subnet/Subnet id = 200.1.2.128
- Direct Broadcast Address = 200.1.2.255
- Total number of IP Addresses = 128
- Range of IP Addresses = [200.1.2.128, 200.1.2.255]
- Total number of hosts that can be configured = 128 - 2 = 126
- Range of Allocated IP Addresses = [200.1.2.129, 200.1.2.254]

Numerical Example

Subnets:

Subnet	Network Address	Broadcast	Host Range
1	192.168.1.0/26	192.168.1.63	.1 – .62
2	192.168.1.64/26	192.168.1.127	.65 – .126
3	192.168.1.128/26	192.168.1.191	.129 – .190

4	192.168.1.192/26	192.168.1.255	.193 – .254
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2. Variable Length Subnetting (VLSM)

Subnets can have **different sizes**.

Used when departments need different numbers of hosts.